

Interacting Multipoles of Rank 3 and 3:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B=R_z^{-13}\cdot\frac{1}{225}\cdot\left.\begin{aligned}&+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\gamma,\delta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\gamma)\cdot\delta(\delta,\epsilon)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\gamma,\epsilon)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\epsilon)\cdot\delta(\gamma,\delta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\beta)\cdot\delta(\epsilon,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\epsilon)\cdot\delta(\beta,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\epsilon)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\zeta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\zeta)\cdot\delta(\beta,\delta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\beta)\cdot\delta(\delta,\epsilon)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\\&+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\delta)\cdot\delta(\beta,\epsilon)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\delta\epsilon\zeta}^B\end{aligned}\right| \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B=R_z^{-13}\cdot\frac{1}{225}\cdot\left\{\begin{array}{l}+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\alpha\beta}^A\cdot\Omega_{\gamma\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\gamma\epsilon}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\alpha\beta}^A\cdot\Omega_{\gamma\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\beta\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\gamma,\gamma)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\gamma\zeta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\alpha\gamma}^A\cdot\Omega_{\beta\beta\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\delta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\delta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\alpha\gamma}^A\cdot\Omega_{\beta\beta\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\epsilon\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\alpha\beta\epsilon}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\alpha\gamma}^A\cdot\Omega_{\beta\beta\zeta}^B\\+105\cdot R_z^4\cdot R_\gamma\cdot R_\zeta\cdot\delta(\alpha,\alpha)\cdot\delta(\beta,\beta)\cdot\Omega_{\alpha\beta\gamma}^A\cdot\Omega_{\beta\beta\zeta}^B\end{array}\right. \quad (2a)$$

[illegible]

[illegible]

$$\begin{aligned}
& \xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned}
& +10395 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot R_\zeta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\delta\epsilon\zeta}^B \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\delta\epsilon\epsilon}^B \\
& -1890 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\gamma \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\delta\delta\epsilon}^B \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\beta \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\gamma\delta\epsilon}^B \\
& -2835 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\beta\delta\epsilon}^B \\
& -945 \cdot R_z^2 \cdot R_\alpha \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\gamma\delta\epsilon}^B \\
& -2835 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\delta\epsilon}^B \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\gamma\delta\epsilon}^B \\
& -945 \cdot R_z^2 \cdot R_\beta \cdot R_\gamma \cdot R_\delta \cdot R_\epsilon \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \Omega_{\beta\delta\epsilon}^B \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\beta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\gamma\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\beta\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\gamma\gamma\delta}^B \\
& +630 \cdot R_z^4 \cdot R_\alpha \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\beta\gamma\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\gamma \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\gamma\gamma\delta}^B \\
& +630 \cdot R_z^4 \cdot R_\beta \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\gamma\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\gamma}^A \cdot \Omega_{\beta\beta\delta}^B \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\delta}^B \\
& +630 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\delta}^B \\
& +315 \cdot R_z^4 \cdot R_\gamma \cdot R_\delta \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\delta}^B \\
& -90 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\gamma}^B \\
& -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\gamma}^B \\
& -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B
\end{aligned} \right] \quad (2d)
\end{aligned}$$

Simplify R:

$$\begin{aligned}
& T_{\alpha\beta\gamma\delta\epsilon\zeta} \Omega_{\alpha\beta\gamma}^A \Omega_{\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned}
& +10395 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^A \cdot \Omega_{zzzz}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^A \cdot \Omega_{z\epsilon\epsilon}^B \\
& -1890 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzzz}^A \cdot \Omega_{zz\delta}^B \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz\gamma}^A \cdot \Omega_{zzz\gamma}^B \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zzz\beta}^A \cdot \Omega_{zzz\beta}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\beta\beta}^A \cdot \Omega_{zzz}^B \\
& -2835 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha\alpha}^A \cdot \Omega_{zzz}^B \\
& -945 \cdot R_z^2 \cdot R_z \cdot R_z \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^A \cdot \Omega_{zzz}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zzz\gamma}^A \cdot \Omega_{\gamma\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zzz\beta}^A \cdot \Omega_{\beta\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\beta\beta}^A \cdot \Omega_{z\gamma\gamma}^B \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\beta\gamma}^A \cdot \Omega_{z\beta\gamma}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{zz\alpha}^A \cdot \Omega_{\alpha\delta\delta}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^A \cdot \Omega_{z\gamma\gamma}^B \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\gamma}^A \cdot \Omega_{z\alpha\gamma}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\alpha}^A \cdot \Omega_{z\beta\beta}^B \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\
& +630 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{zz\beta}^B \\
& +315 \cdot R_z^4 \cdot R_z \cdot R_z \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{zz\alpha}^B \\
& -90 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\gamma}^B \\
& -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\gamma}^B \\
& -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B
\end{aligned} \right] \quad (3a)
\end{aligned}$$

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-2835 \cdot R_z^6 \cdot \Omega_{zz\gamma}^A \cdot \Omega_{zz\gamma}^B - 2835 \cdot R_z^6 \cdot \Omega_{zz\beta}^A \cdot \Omega_{zz\beta}^B \\ &-2835 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B + 315 \cdot R_z^6 \cdot \Omega_{zz\gamma}^A \cdot \Omega_{\gamma\delta\delta}^B \\ &+ 315 \cdot R_z^6 \cdot \Omega_{zz\beta}^A \cdot \Omega_{\beta\delta\delta}^B + 630 \cdot R_z^6 \cdot \Omega_{z\beta\gamma}^A \cdot \Omega_{z\beta\gamma}^B \\ &+ 315 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{\alpha\delta\delta}^B + 630 \cdot R_z^6 \cdot \Omega_{z\alpha\gamma}^A \cdot \Omega_{z\alpha\gamma}^B \\ &+ 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B + 630 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{zz\beta}^B \\ &+ 315 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{zz\alpha}^B - 90 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\gamma}^B \\ &\quad - 45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\gamma}^B \\ &\quad - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{aligned} \right] \quad (3b)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{array}{l} -2835 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B - 2835 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B \\ -2835 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B + 315 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{\alpha\beta\beta}^B \\ +315 \cdot R_z^6 \cdot \Omega_{zz\beta}^A \cdot \Omega_{\alpha\alpha\beta}^B + 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ +315 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{\alpha\beta\beta}^B + 630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ +630 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B + 630 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{z\beta\beta}^B \\ +315 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{zz\alpha}^B - 90 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\gamma}^B \\ -45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\gamma}^B \\ -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{array} \right] \quad (3c)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{array}{l} -8505 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{zz\alpha}^B + 630 \cdot R_z^6 \cdot \Omega_{zz\alpha}^A \cdot \Omega_{\alpha\beta}^B \\ + 315 \cdot R_z^6 \cdot \Omega_{zz\beta}^A \cdot \Omega_{\alpha\beta}^B + 1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ + 630 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{zz\beta}^B + 315 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{zz\alpha}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{\alpha\alpha\beta}^A \cdot \Omega_{\beta\gamma\gamma}^B \\ - 45 \cdot R_z^6 \cdot \Omega_{\alpha\beta\beta}^A \cdot \Omega_{\alpha\gamma\gamma}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{array} \right] \quad (3d)$$

Simplify Oktopoles:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 630 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{y\beta\beta}^B \\ &+ 315 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{y\alpha\alpha}^B + 1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ &+ 630 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{yzz}^B + 315 \cdot R_z^6 \cdot \Omega_{y\beta\beta}^A \cdot \Omega_{yzz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{y\gamma\gamma}^B - 45 \cdot R_z^6 \cdot \Omega_{y\beta\beta}^A \cdot \Omega_{y\gamma\gamma}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{aligned} \right] \quad (4a)$$

$$\xrightarrow{\text{reduced indices}} R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{array}{c} -8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 630 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{y\alpha\alpha}^B \\ + 315 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{y\alpha\alpha}^B + 1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ + 630 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{yzz}^B + 315 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{yzz}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{y\beta\beta}^B - 45 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{y\beta\beta}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{array} \right] \quad (4b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \begin{bmatrix} -8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 945 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{y\alpha\alpha}^B \\ +1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B + 945 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{yzz}^B \\ -135 \cdot R_z^6 \cdot \Omega_{y\alpha\alpha}^A \cdot \Omega_{y\beta\beta}^B - 90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{bmatrix} \quad (4c)$$

Exploit Traceless Conditions:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{225} \cdot \begin{bmatrix} -8505 \cdot R_z^6 \cdot \Omega_{yz}^A \cdot \Omega_{yz}^B + 1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{bmatrix} \quad (5a)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \begin{bmatrix} -8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ +1890 \cdot R_z^6 \cdot \Omega_{z\alpha\beta}^A \cdot \Omega_{z\alpha\beta}^B \\ -90 \cdot R_z^6 \cdot \Omega_{\alpha\beta\gamma}^A \cdot \Omega_{\alpha\beta\gamma}^B \end{bmatrix} \quad (5b)$$

Multiply Out Sum:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B = R_z^{-13} \cdot \frac{1}{225} \cdot \left[\begin{aligned} &-8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 1890 \cdot R_z^6 \cdot \Omega_{xxz}^A \cdot \Omega_{xxz}^B \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B + 1890 \cdot R_z^6 \cdot \Omega_{xzz}^A \cdot \Omega_{xzz}^B \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{xyx}^A \cdot \Omega_{xyx}^B + 1890 \cdot R_z^6 \cdot \Omega_{yyz}^A \cdot \Omega_{yyz}^B \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 1890 \cdot R_z^6 \cdot \Omega_{xzz}^A \cdot \Omega_{xzz}^B \\ &+ 1890 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 1890 \cdot R_z^6 \cdot \Omega_{zzz}^A \cdot \Omega_{zzz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xxx}^A \cdot \Omega_{xxx}^B - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xxz}^A \cdot \Omega_{xxz}^B - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xyy}^A \cdot \Omega_{xyy}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xxz}^A \cdot \Omega_{xxz}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xzz}^A \cdot \Omega_{xzz}^B - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xyy}^A \cdot \Omega_{xyy}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xyy}^A \cdot \Omega_{xyy}^B - 90 \cdot R_z^6 \cdot \Omega_{yyy}^A \cdot \Omega_{yyy}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{yyz}^A \cdot \Omega_{yyz}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{yyz}^A \cdot \Omega_{yyz}^B - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xxz}^A \cdot \Omega_{xxz}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xzz}^A \cdot \Omega_{xzz}^B - 90 \cdot R_z^6 \cdot \Omega_{xyz}^A \cdot \Omega_{xyz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{yyz}^A \cdot \Omega_{yyz}^B - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{xzz}^A \cdot \Omega_{xzz}^B - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ &- 90 \cdot R_z^6 \cdot \Omega_{zzz}^A \cdot \Omega_{zzz}^B \end{aligned} \right] \quad (6a)$$

$$\xrightarrow{\text{cleaned up}} R_z^{-13} \cdot \frac{1}{225} \cdot \begin{bmatrix} -8505 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B + 1890 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ + 1890 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B - 90 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{yyy}^A \cdot \Omega_{yyy}^B - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B - 90 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \end{bmatrix} \quad (6b)$$

$$\xrightarrow{\text{added up}} R_z^{-13} \cdot \frac{1}{225} \cdot \begin{bmatrix} -4995 \cdot R_z^6 \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ -270 \cdot R_z^6 \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ -90 \cdot R_z^6 \cdot \Omega_{yyy}^A \cdot \Omega_{yyy}^B \end{bmatrix} \quad (6c)$$

Combining everything:

$$T_{\alpha\beta\gamma\delta\epsilon\zeta}\Omega_{\alpha\beta\gamma}^A\Omega_{\delta\epsilon\zeta}^B = \begin{bmatrix} -\frac{111}{5} \cdot R_z^{-7} \cdot \Omega_{yzz}^A \cdot \Omega_{yzz}^B \\ -\frac{6}{5} \cdot R_z^{-7} \cdot \Omega_{xxy}^A \cdot \Omega_{xxy}^B \\ -\frac{2}{5} \cdot R_z^{-7} \cdot \Omega_{yyy}^A \cdot \Omega_{yyy}^B \end{bmatrix} \quad (7)$$